

Fernando Campos Silva Dal' Maria

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Summary

Computer Science student at PUC Minas with a GPA of **9.475/10.0**, focused on **computer graphics**, **artificial intelligence**, and high-performance computing. Experienced in building graph and automata visualization tools, GPU-accelerated C++ machine learning frameworks, and AI-driven automation solutions as an independent contractor. Fluent in English, with strong foundations in algorithms, machine learning, model interpretability, and interactive TypeScript applications.

Education

Bachelor of Science in Computer Science

PUC Minas — Pontifícia Universidade Católica de Minas Gerais

Belo Horizonte, Brazil

2022 – Present

- GPA: **9.475 / 10.0**, consistently ranked among top students in the cohort
- Academic Distinctions: Dean's List / Laurea recognition for outstanding academic performance
- Teaching Assistant (Monitor): Histology, Graph Theory, Computability Theory, and Database Systems

Medicine (incomplete, transferred to Computer Science)

Faculdade Ciências Médicas de Minas Gerais

Belo Horizonte, Brazil

2020 – 2022

- Two years of training in anatomy, physiology, biochemistry, histology, and pathology

Experience

Software Development Intern

Tribunal de Contas do Estado de Minas Gerais (TCE-MG)

Belo Horizonte, Brazil

Apr 2026 – Present

- Support software development and technology initiatives within the State Court of Accounts of Minas Gerais, contributing to internal systems and public-sector digital transformation projects

Freelance Data Analyst & AI Automation Engineer

Self-employed (Independent Contractor)

Brazil

2023 – Present

- Designed and implemented an internal management system for a furniture store, supporting operational workflows, data organization, and process automation
- Developed a relationship-graph analysis system to identify potential nepotism patterns by modeling entities and connections as graph structures
- Collected, structured, and analyzed patient data for a research project involving participants from the UFMG university hospital, supporting exploratory analysis and reporting
- Built a PyTorch-based automation system for soybean crop disease identification, designing and training custom deep learning models for image-based classification

Undergraduate Researcher: CNPq Scholarship & Teaching Assistant (Monitor)

PUC Minas — LICAP Lab

Belo Horizonte, Brazil

2023 – 2025

- Published at **IHC 2024** (XXIII Brazilian Symposium on Human Factors in Computing Systems, ACM): “*Persuasive Technology for Managing the Routine of Families with Children and Adolescents with ADHD: A User-Centered Approach*”, received an **Honorable Mention** [doi:10.1145/3702038.3702101](https://doi.org/10.1145/3702038.3702101). Complementary publication in the **Journal of the Brazilian Computer Society (JBCS)**: [doi:10.5753/jbcs.2026.5670](https://doi.org/10.5753/jbcs.2026.5670)
- Current independent research proposal: interpretability of deep learning and foundation models using the **kernel trick** and custom-designed kernels to measure geometric similarity across each layer's representation space, enabling mechanistic understanding of model decisions
- Mentored undergraduate students in Graph Theory, Computability & Automata Theory, Database Systems, and Histology
- Conducted office hours, graded assignments, and produced supplementary educational materials

Personal Projects

Selected personal projects demonstrating knowledge and proficiency in computer graphics and artificial intelligence.

Pineapple: C++ Machine Learning Framework

 [FernandoCsm-Knight/Pineapple](#)

- Built a PyTorch/TensorFlow-inspired deep learning framework from scratch in C++, featuring automatic differentiation, tensor operations, and a modular layer API
- Implements **parallel computing** on both CPU (multithreading) and GPU (CUDA), achieving significant performance gains over naive implementations
- Covers neural network architectures, optimizers, and loss functions, designed for research and performance-critical applications

Graphic Paint: Graph & Automaton Visual Editor

 [Graphic-Paint](#) ·  [Live Demo](#)

- A web-based drawing application (Excalidraw-inspired, built with TypeScript) for constructing directed/undirected graphs and finite automata (DFA, NFA, ε -NFA, Pushdown Automata)
- Supports **input simulation on automata** and **graph algorithm execution** (BFS, DFS, shortest path, MST, etc.) directly within the canvas
- Deployed on **Vercel**; includes user authentication and cloud data persistence

Stellar Classification

 [Stellar-Classification](#)

- Applied classical and modern ML classifiers (Random Forest, SVM, Neural Networks) to astronomical datasets for automated star/galaxy/quasar classification

Certifications & Awards

- **Microsoft Azure AI Fundamentals (AI-900)** — Microsoft
- **Red Hat Certified System Administrator II (RHCSA)** — Red Hat
- **Honorable Mention** — Academic Congress / Scientific Initiation (PUC Minas)
- **Academic Distinction** — Dean's recognition for outstanding GPA and performance

Technical Skills

Languages	C, C++, Python, Java, TypeScript, JavaScript, Rust
ML / AI	Deep Learning, Kernel Methods, Model Interpretability, Computer Vision, NLP
Frameworks	PyTorch, TensorFlow, scikit-learn, React, Node.js
Parallel Computing	CUDA, Multithreading (C++), OpenMP, PThreads
Cloud & DevOps	Microsoft Azure, Linux (RHEL/Ubuntu), Vercel, Git, Docker
Databases	PostgreSQL, MySQL, MongoDB, Firebase
Languages (Human)	Portuguese (Native), English (Fluent, professional working proficiency)